

EXPERIMENT – 0 GO FURTHER – 2

PROJECT NAME

7 Segment Display Interfacing with Arduino UNO

OBJECTIVE / PROBLEM STATEMENT

The objective of this project is to interface a common cathode 7-segment display with Arduino UNO and display numbers sequentially.

This experiment helps in understanding:

- 7-segment display interfacing
- Digital output control
- Number display logic
- Multi-pin control using Arduino

COMPONENTS USED

Component	Quantity
Arduino UNO	1
7 Segment Display (Common Cathode)	1
220Ω Resistors	7
Jumper Wires	Required
Breadboard	1

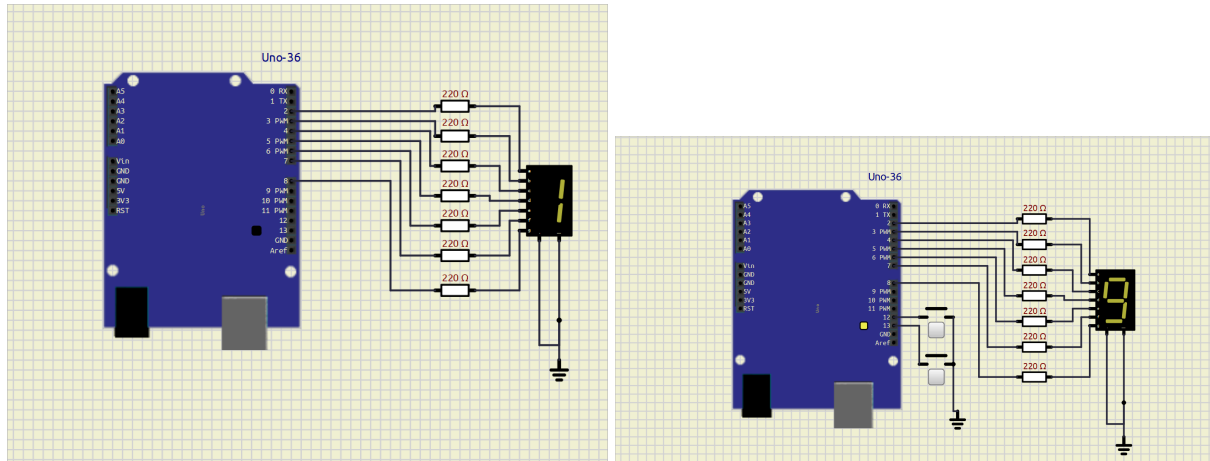
PIN CONNECTIONS

7 Segment Pin	Arduino Pin
a	D2
b	D3
c	D4
d	D5
e	D6
f	D7
g	D8

SOFTWARE USED

- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM



MAIN ARDUINO CODE

```
// 7 Segment Display with Arduino UNO  
// Common Cathode Display
```

```
int a = 2;  
int b = 3;  
int c = 4;  
int d = 5;  
int e = 6;  
int f = 7;  
int g = 8;
```

```
void setup() {
```

```
  pinMode(a, OUTPUT);  
  pinMode(b, OUTPUT);  
  pinMode(c, OUTPUT);  
  pinMode(d, OUTPUT);  
  pinMode(e, OUTPUT);  
  pinMode(f, OUTPUT);  
  pinMode(g, OUTPUT);
```

```
}
```

```
void loop() {
```

```
  // Display 0  
  digitalWrite(a, HIGH);  
  digitalWrite(b, HIGH);
```

```
digitalWrite(c, HIGH);
digitalWrite(d, HIGH);
digitalWrite(e, HIGH);
digitalWrite(f, HIGH);
digitalWrite(g, LOW);

delay(1000);

// Display 1
digitalWrite(a, LOW);
digitalWrite(b, HIGH);
digitalWrite(c, HIGH);
digitalWrite(d, LOW);
digitalWrite(e, LOW);
digitalWrite(f, LOW);
digitalWrite(g, LOW);

delay(1000);

// Display 2
digitalWrite(a, HIGH);
digitalWrite(b, HIGH);
digitalWrite(c, LOW);
digitalWrite(d, HIGH);
digitalWrite(e, HIGH);
digitalWrite(f, LOW);
digitalWrite(g, HIGH);

delay(1000);

// Display 3
digitalWrite(a, HIGH);
digitalWrite(b, HIGH);
digitalWrite(c, HIGH);
digitalWrite(d, HIGH);
digitalWrite(e, LOW);
digitalWrite(f, LOW);
digitalWrite(g, HIGH);

delay(1000);

}
```

WORKING PRINCIPLE

1. The 7-segment display segments are connected to Arduino digital pins D2 to D8.
2. Each segment is controlled individually using `digitalWrite()`.
3. Depending on which segments are turned ON or OFF, different numbers are displayed.
4. The Arduino sequentially displays:

- 0
- 1
- 2
- 3

5. Each number remains visible for 1 second using `delay(1000)`.

6. The sequence repeats continuously.

WHAT I LEARN

- Interfacing a 7-segment display with Arduino UNO
 - Common cathode display working principle
 - Using multiple digital output pins
 - Number display logic using segments
 - Arduino digital output voltage is approximately 5V
 - Recommended current per pin is 20mA
 - Typical LED segment current is around 10mA to 20mA
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ADVANTAGES

- Simple number display system
 - Easy to understand segment logic
 - Useful for beginners
 - Low-cost implementation
-

APPLICATIONS

- Digital counters
 - Timer displays
 - Calculator displays
 - Embedded display systems
 - Electronic meters
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CONCLUSION

The 7-segment display was successfully interfaced with Arduino UNO.

The display sequentially showed numbers from 0 to 3 by controlling individual segments through Arduino digital pins. This project helped in understanding display interfacing and segment control logic.

REFERENCES

1. Arduino Official Website
<https://www.arduino.cc/>

2. Arduino UNO Documentation

<https://docs.arduino.cc/hardware/uno-rev3/>

3. 7 Segment Display Basics

<https://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html>

4. Arduino Tutorials

<https://www.arduino.cc/en/Tutorial/HomePage>